

Visvesvaraya Technological University Belagavi



A Mini Project Report

on

VISIBLE LIGHT COMMUNICATION (Li-Fi) BASED DATA TRANSMISSION SYSTEM

Submitted by

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*In partial fulfilment for the award of the
degree of*

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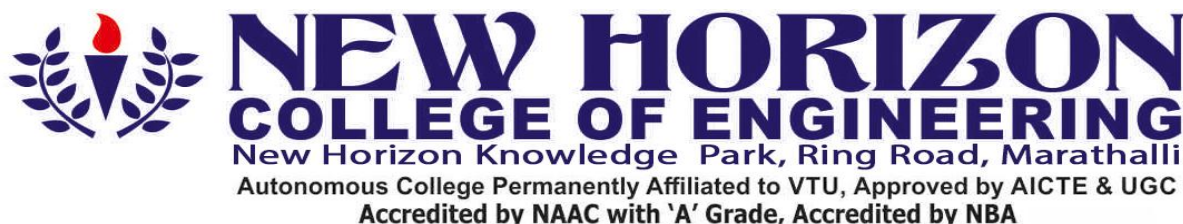
IN

ELECTRONICS AND COMMUNICATION ENGINEERING



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Certified that the Mini project entitled “**VISIBLE LIGHT COMMUNICATION (LI-FI) BASED DATA TRANSMISSION SYSTEM** ” is carried out by **Mr. APPU RATHOD** bearing **USN:1NH24EC015** and **Mr. DARSHAN** bearing **USN:1NH24EC028**, bonafide students of NHCE, Bengaluru in partial fulfilment for the award of Bachelor of Engineering in Electronics and Communication of the Visvesvaraya Technological University, Belagavi during the year 2025-26. It is certified that all corrections and suggestions indicated for Internal Assessment have been incorporated in the report deposited in the department library. The mini project report has been approved as it satisfies the academic requirements in respect of the mini project work prescribed for the said degree.

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ABSTRACT

Li-Fi (Light Fidelity) is a modern wireless communication technology that uses visible light to transmit data instead of the radio frequency signals used in conventional Wi-Fi systems. The main aim of this project is to design and implement a simple Li-Fi data transfer system using Arduino UNO, a 16x2 LCD display, an LDR sensor, a breadboard, a 1k ohm resistor, and jumper wires.

This project employs Arduino Uno in sending data in the form of signals that connect to the LED on the transmitting side. In this process, Arduino regulates the blinks of LED during high transmission of signals. The blinks are not noticeable to the naked eye but they are able to transmit information via a free space. The sensor LED is used to detect the variations in light intensity which are not seen by human beings. On the other hand, the receiver uses an LDR sensor which detects the variations in the light being transmitted. After detection, the signal is then processed using Arduino Uno receiver connection and the information is displayed on the LCD screen.

This Project Demonstrates How visible light can be use as wireless communication in a simple and cost effective purpose , and it will used to compare to modern traditional wireless communication systems, Li-Fi Technology offers us to such as higher security , low interference, energy efficiency, and utilization of a spectrum. This technology can be useful in places where radio frequency communication is restricted, such as hospitals and wireless communication industries.

This project is simple, Cost effective, and energy efficient and making it for educational purpose, and short range wireless communication applications . This connections also highlights the future scope of Li-Fi technology as an modern wireless short distance communication systems.

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CHAPTER 1

INTRODUCTION

In a modern world new generation, the wireless Communication has become the most important part of their daily life. Most wireless communications systems are mostly radio frequency signals to transmitting the data through wireless communication such as Wi-Fi, Bluetooth, and mobile networks. Now as the number of wireless devices increases, the radio frequency become crowded, it will lead issues like limited bandwidth, signal interference, security issues, , reducing the data network security concerns. To overcome these limitations , the IEEE and researchers search the alternative way and introduced the other wireless communication called as light fidelity (Li- Fi).

Li-Fi is a wireless communication technology that used the visible light to transmit the medium range of data like texts and etc. This Li-Fi concept was introduced by Harald Haas. Who tried to demonstrate the LED Light can be used not only for illuminations but also for data communication. In Li-Fi system Data transmits by rapidly through by switching an LED light by ON and OFF at high speed. These Changes are too fast it will not visible for humans eye to detect but it can be sensed by light sensitive devices such as photodiodes or LDR sensors.

This Project mainly focuses on the design and implementation of a Li-Fi based data transfer system using Arduino UNO , LCD Display, LDR Sensor, Resistor, Breadboard, and Jumper Wires. In the transmitter section and received using visible light communication.

In a praposed system , a Li-Fi mobile application is used for generate and transmit the light signals through LED source. The Transmitted light falls on the LDR sensor at the receiver side. The LDR sensor it will detect the changes in light source and converts them into electrical signals. These signals are processed using Arduino UNO, and the received output is displayed on the LCD display.

The developed system demonstrates short distance data communication through visible light and highlights the future scope of Li-Fi technology In a modern communication systems.

CHAPTER 2

LITERATURE REVIEW

2.1 Li-Fi Data Transmission Through visible Light – HARALD HAAS (2011)

2.1.1 : Methodology : Harald Haas introduced the concept of Li-Fi technology using visible LED light for wireless communication. The system used LED light modulation to transmit digital data through visible light communication instead of radio frequency signals .

2.1.2 : Outcome : The research successfully demonstrated for wireless data communication system using LED light and he proved that visible light communication can achieve speed data transfer and better bandwidth than radio frequency .

2.1.3 : Limitation : The system required direct line of sight communication and the performance was affected by ambient light interference and physical obstacles blocking light path.

2.2 Visible Light Communication : A Potential Solution – IEEE Researchers (2015)

2.2.1 : Methodology : The researchers Implemented a visible light Communication system using LED transmitters and photo detectors receivers for secure indoor wireless communication.

2.2.2 : Outcome : The developed system achieved high data transmission rates and provided secure communication in indoor environments with reduced electromagnetic interference.

2.2.3 : Limitations : The communication range was limited and the system could not function properly when the light path was interrupted or blocked .

2.3 Design and Implementation of Li-Fi System – S. Rajagopal et al. (2014)

2.3.1 : Methodology : The authors designed a Li-Fi communication using LED transmitters and photo diode receivers for transmitting audio and digital data signals through visible light.

2.3.2 : Outcome : The system successfully transmitted data and audio signals over short distances and demonstrated the practical implementation of Li-Fi technology.

2.3.3 : Limitation : The system had a short transmission range and signal quality was affected by external light sources and environmental disturbances.

2.4 Wireless Communication Using Visible Light Technology – D.O' Brien et al.(2015)

2.4.1: Methodology : The authors developed a wireless communication system using visible light technology with LED transmitters and optical receivers for indoor data communication applications.

2.4.2 : Outcome : The research demonstrated efficient wireless communication through visible light and achieved stable short range data transmission with improved bandwidth efficiency.

2.4.3 : Limitation : The system performance depended heavily on proper alignment between transmitter and receiver and was affected by external interference.

CHAPTER 3

EXISTING SYSTEM

Out in the open, most current wireless setups rely on radio-based methods – think Wi-Fi, Bluetooth, or cell networks – to send data back and forth. Instead of wires, they push signals using invisible radio waves that travel through space. We'll find them running things at home, inside factories, across workspaces, city-wide connections.

Even with its ease of connection, Wi-Fi struggles now that so many people and gadgets rely on it. More users mean tighter bandwidth, overloaded networks, clashing signals, weak protection. When too many devices crowd one space, they share the same airwaves – slowing data flow. Performance dips where traffic peaks.

Where radios might disrupt vital machines – like inside planes or medical centers sending signals through airwaves becomes risky. Because of this, another kind of invisible link must step in, one that moves information safely without causing harm. Sometimes light carries the message instead, slipping through space without upsetting nearby devices. Safety often means avoiding familiar methods when silence and precision matter most.

LIMITATIONS OF CURRENT SYSTEMS

1. Most Wi-Fi setups rely on radio waves, a type of signal that's already packed full. Space in this range runs short because so many devices crowd it.
2. Signal interference occurs when multiple wireless devices operate in the same frequency range.
3. Fewer gadgets handle radio signals well when power use climbs. Signals at high strength sometimes disrupt nearby electronics.
4. Fences around signals mean little when waves slip through walls, making grabs by outsiders quite possible.
5. Funny how a stray radio signal can mess things up, especially when weather adds its own twist. Wires hum under pressure while unseen waves crash into each other out of nowhere.

CHAPTER 4

PROPOSED SOLUTION

A small torch on your phone beams information through light. Instead of radio waves, it sends signals you can see. Light hits a special sensor called an LDR. That part catches what the phone flashes out. The whole setup swaps invisible frequencies for glowing pulses. What usually travels through air now moves through brightness changes. A basic bulb becomes the messenger instead of antennas.

A smartphone's beam of light carries words across space in this setup. When someone types a message into the app named Li-Fi, it kicks off a sequence where letters turn into flickers. Instead of sending radio waves, the phone flashes its torch quickly – so fast the eye barely notices. Hidden inside those bursts are ones and zeros shaped by brightness changes. Information rides on rays, moving through air like code written in light. What looks like blinking holds full messages meant for another device.

On the receiving end, a small light-sensing resistor picks up shifts in brightness from the phone's torch. As light levels shift, this sensor turns those variations into tiny electric pulses. These pulses travel to a compact control board that makes sense of the incoming patterns. Once interpreted, the decoded information appears line by line on a clear screen nearby.

Light carries data in this setup, offering a straightforward way to communicate without wires. Because it relies on illumination rather than radio waves, unwanted signal overlap drops off sharply. Security improves naturally – someone must be in the room to intercept the stream. This version works best across small spaces, fitting neatly into nearby device exchanges. Seeing how light transmits information becomes clear through hands-on use of such designs.

CHAPTER 5

HARDWARE AND SOFTWARE COMPONENTS

5.1 : HARDWARE COMPONENTS :

1. Arduino UNO Microcontroller :



Fig 5.1 : Arduino UNO

Specifications: Atmega328P microcontroller , Operating Voltage: 5V , Input Voltage: 7V–12V , 14 Digital I/O Pins , 6 Analog Input Pins , 32 KB Flash Memory, Clock Speed: 16 MHz

Role:

Light hits the sensor, then the Arduino UNO takes over. This board reads what comes in through the LDR. After that, it turns flashes into readable text. The result shows up line by line on the screen. One piece talks to another thanks to its guidance. Messages move from brightness to words because it manages each step. Parts listen and respond only when it leads them.

2. LDR Sensor :



Fig 5.2 : LDR Sensor

Specifications:

Light Dependent Resistor (LDR) ; Operating Voltage : 3V–5V ;High resistance in darkness ; Low resistance in presence of light shifts? Reaction happens quick. Sudden brightness brings instant adjustment. Changes in glow meet immediate reaction. When illumination jumps, response follows without delay. Brightness alters, system reacts right away

Role:

Light hits the LDR sensor when the phone's flashlight blinks using a special app. Though small, it plays a key role by catching those flickers in brightness. As levels shift, its internal resistance shifts too – no sudden jumps, just smooth steps. This change feeds an electric signal into the Arduino UNO instead of staying trapped in light form. Processing begins once that data arrives quietly at the board.

3. 16*2 LCD Display :



Fig 5.3: 16*2 LCD Display

Specifications:

Display Type: 16x2 LCD , Operating Voltage: 5V, 16 Pins Interface ,Each row shows a screen ,Two display rows

Role:

Now showing on the screen: whatever info comes in via the light-based link. Once the tiny brain chip works out what the light sensor caught, it pushes the cleaned-up words over to the little window so you can see them right there.

4. 1k Ohm Resistor :



Fig 5.4 : Resistor

Specifications: Resistance Value: $1k\Omega$,Fixed resistor, Power Rating: 0.25W
Tolerance: $\pm 5\%$

Role:

A small part controls how much electricity moves through the system. This device shapes signals while keeping power levels safe. Instead of letting surges damage parts, it slows down the flow. Stability in the light-based receiver comes from its steady influence.

5. Half Bread Board

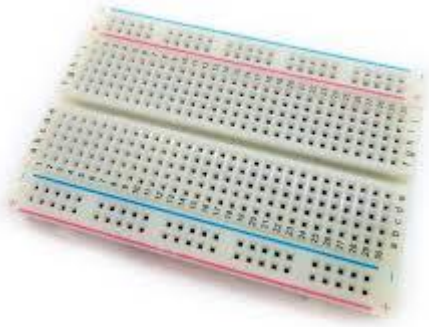


Fig 5.5 : Half Bread Board

Specifications:

Solderless breadboard ,Reusable circuit platform,Internal conductive strips
Supports temporary circuit connections

Role:

A small part controls how much electricity moves through the system. This device shapes signals while keeping power levels safe. Instead of letting surges damage parts, it slows down the flow. Stability in the light-based receiver comes from its steady influence.

5.2 SOFTWARE COMPONENTS

1. Arduino IDE

Role: Most folks turn to the Arduino IDE when they need to craft code for the UNO chip. Writing lines happens in a clean workspace that feels straightforward. Code gets checked right inside the app before moving forward. Once ready, it travels straight into the microcontroller without fuss. Hooking up an LDR sensor becomes easier thanks to built-in tools. The screen part? An LCD talks smoothly through guidance baked into the software. Testing how light carries data uses the same interface too. Serial links open directly so messages can pass back and forth.

2. Li-Fi Mobile Application

Role: Inside the setup, the Li-Fi phone app works like a sender. When someone types a message, instead of staying as words on screen, it turns that text into flashes using the phone's torchlight – fast enough to stream data. A flickering beam carries these pieces across space. At the far end, an LDR waits quietly, catching each glow and translating brightness back into readable letters seen once more.

BLOCK DIAGRAM

BLOCK DIAGRAM – LI-FI DATA TRANSFER SYSTEM

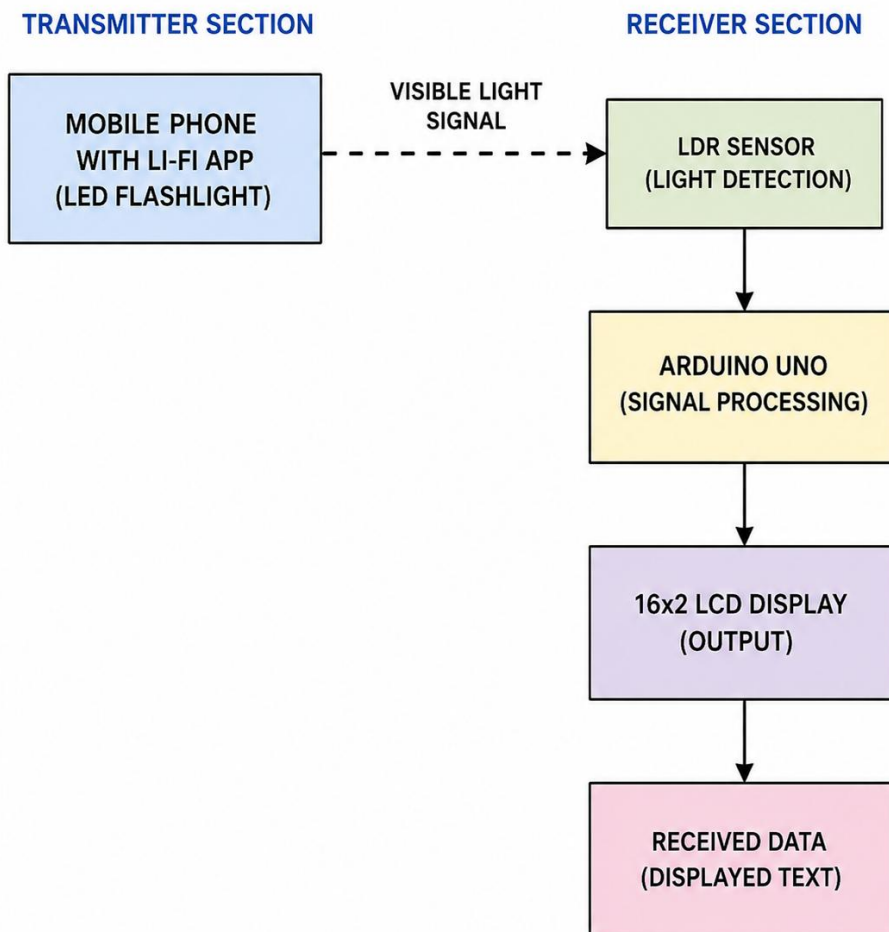


Fig 5.6 : Block Diagram Of The Project

CHAPTER 6

ADVANTAGES AND APPLICATIONS

6.1 ADVANTAGES

- Light Fidelity Technology uses visible light for wireless communication as opposed to radio frequency communication.
- This technology offers high levels of security because the visible light cannot penetrate walls hence no access by unauthorized personnel.
- It ensures minimum electromagnetic interference when compared to other Wi-Fi systems.
- The implementation of this technology is simple, affordable, and involves cheap electronic components.
- The communication process consumes minimal energy because it uses LED lights for both lighting and wireless communication.
- Visible light-based communication technology enables faster and secure communication at short distances.
- It can be applied in areas where radio frequency communication is not permitted due to safety concerns like hospitals, airplanes, and military locations.
- Wireless communication does not involve internet or radio waves, making communication effective and efficient.
- Power consumption is relatively low in this system and few electronic parts are involved.
- Light Fidelity Technology uses a wide bandwidth that enhances wireless communication and eliminates network congestion.

6.2 APPLICATIONS

- **Hospitals :**
The Li-Fi technology is applied in hospitals to ensure no interference takes place while maintaining communication with safety.
- **Aircraft Communication :**
The Li-Fi technology is employed in aircraft to enable wireless communication without interfering with the navigation system.
- **Underwater Communication :**
The technology is employed in facilitating communication under water where radio waves fail to travel.
- **Smart Classrooms :**
The technology is employed in classrooms as means of enabling wireless data sharing using LED light sources.
- **Military Applications :**
The use of Li-Fi technology in military communication ensures the security of the message since light does not pass through the walls.
- **Traffic Management :**
Traffic signals and vehicle communication system utilize Li-Fi to enable traffic updates.
- **Smart Homes :**
The technology is applicable in homes to facilitate home automation.
- **Industrial Applications :**
The technology is important in industries where radio frequency communication causes interference.
- **Internet Communication :**
The technology may be employed to substitute Wi-Fi technology in high speed internet communication.

CHAPTER 7

RESULTS

The Li-Fi data transfer system that was discussed above has been designed using various components: Arduino UNO, LDR sensor, LCD display and a mobile flash light + Li-Fi application software. In this case we have successfully transferred text data through a visible light communications method.

The experiments shown above as well as a number of other experiments indicated that the typing of text by the Li-Fi application was being converted into signals from the flash light. The LDR detected variations in the intensity of the light and then transmitted the encoded signal to the Arduino UNO. Upon receiving this signal, the Arduino processed the signal and displayed the text on the LCD display.

It was noted that the quality of the wireless communication received from the Li-Fi flash light system increased greatly when there was a clear line of sight (no obstruction to the transfer of signals) and there were not any sources of light interfering with the transfer of signals. When obstructions (like people) were blocking the transfer, the quality of the wireless communication dropped substantially.

Light has been successfully demonstrated to be an alternative to radio waves for use in communications (wireless communications). This implementation is very simple, low cost and energy efficient.

In conclusion, the project met all of its objectives.

FINAL VIEW

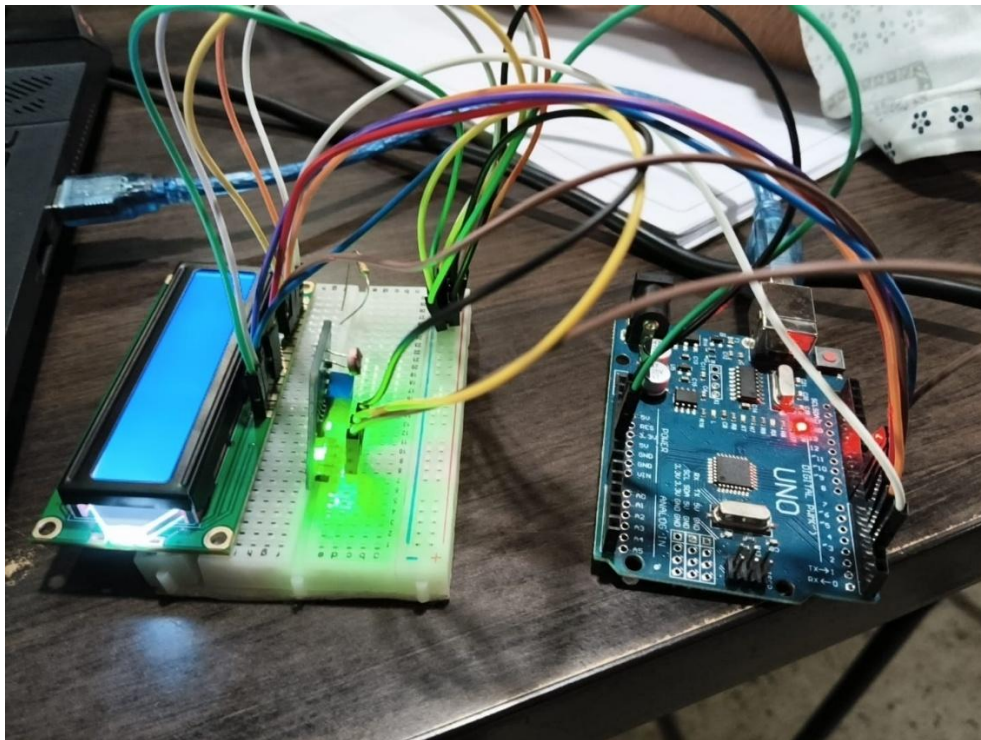


Fig 7.1 : Final View Of the project

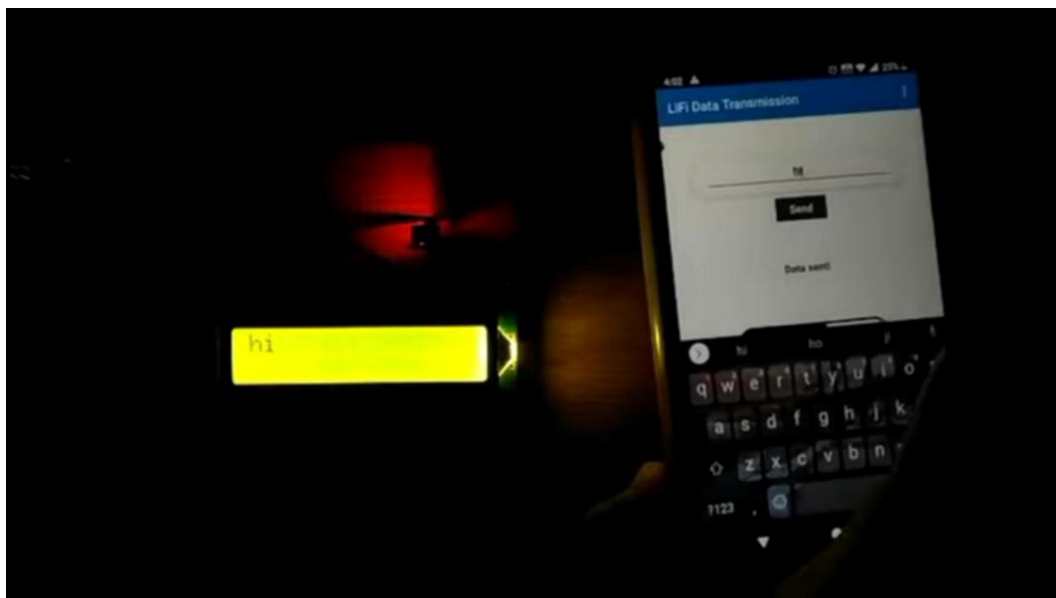


Fig 7.2 : Output Of the Project

CHAPTER 8

CONCLUSION

Li-Fi Data Transfer System has been successfully developed using components such as Arduino UNO, LDR sensor, 16x2 LCD screen, and a mobile flashlight equipped with Li-Fi application. Through the application, the transmitted and received text messages have been transferred using light waves.

Through the mobile Li-Fi application, the entered data was converted into flashlight messages that were then detected by the LDR sensor and sent to the Arduino UNO for further analysis. The received data was then displayed using the 16x2 LCD screen.

The system could transmit and receive data within a short distance when there is proper alignment of both the LDR sensor and mobile flashlight. Through the development, the project was able to demonstrate effective wireless data transfer using light instead of RF signals.

Li-Fi technology can therefore be applied in various applications for wireless data transfer in a cost-effective way.

CHAPTER 9

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APPENDIX

ARDUINO UNO CODE :

```
#include <LiquidCrystal.h>

// LCD pins: RS, EN, D4, D5, D6, D7
LiquidCrystal lcd(2, 3, 4, 5, 6, 7);

// LDR sensor pin
int ldr = 8;

int val = 0;

void setup()
{
  Serial.begin(9600);

  pinMode(ldr, INPUT);

  lcd.begin(16, 2);

  lcd.clear();

  lcd.setCursor(0,0);
  lcd.print("LiFi Project");

  delay(2000);

  lcd.clear();
}

void loop()
{
  val = digitalRead(ldr);

  lcd.setCursor(0,0);
  lcd.print("LiFi Receiver");

  lcd.setCursor(0,1);

  if(val == HIGH)
  {
```

```
    lcd.print("LIGHT DETECTED");  
  }  
  else  
  {  
    lcd.print("NO LIGHT  ");  
  }  
  
  delay(300);  
}
```